

VARAGA HAGHOUBIANS

JUNIOR DATA SCIENTIST & DATA ANALYST | MACHINE LEARNING

Statistical modeling · Industrial time series · Python & ML

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Available now, full-time | Internships & junior roles | Open to relocation in Italy

PROFESSIONAL SUMMARY

• **Industrial Engineering graduate** and MSc student in Stochastics and Data Science at the University of Turin (expected April 2027). **Data Science internship** experience at Eurix using **real industrial data**.

• Developed a **Python pipeline** for 30-minute HVAC/BMS sensor and weather time series; engineered time-series features and trained two seasonal **LSTM autoencoders** tuned with **Optuna** (test reconstruction R^2: 0.972 summer; 0.950 winter).

• Projects in customer **clustering** and **classification**, industrial analytics, and **demand forecasting** on synthetic data. Seeking internships and junior positions in data science, machine learning and data analysis.

TECHNICAL SKILLS

Tools: Python · SQL · Advanced Excel · Git/GitHub

Machine learning: scikit-learn, TensorFlow/Keras, LSTM autoencoders, clustering, classification, PCA, anomaly detection

Statistics: exploratory analysis, feature engineering, statistical tests, time-series validation, R^2 and MAPE

Applied domains: industrial time series and anomaly detection; production and inventory KPIs (internships); demand forecasting (portfolio)

INTERNSHIP EXPERIENCE

Data Science Intern (curricular internship, MSc thesis) ***Apr 2025 - Apr 2026***

Eurix, Turin, Italy

• Developed a reproducible **Python pipeline** to clean and align real HVAC/BMS sensor and weather data into 30-minute time series, handling missing values, outliers and timestamps.

• Engineered time-series features and trained two seasonal **LSTM autoencoders** tuned with **Optuna**; test-set reconstruction **R^2**: 0.972 (summer) and 0.950 (winter).

• Produced potential-anomaly indicators and **technical reports in Italian and English**; flags do not constitute confirmed faults.

Industrial Engineering Intern ***Aug 2020 - Aug 2021***

Arvest, Tehran, Iran

• Digitized paper registers for over 50 material and item types in **Excel**; directly monitored consumption, stock levels, and **safety stock** for raw-material groups A, B, and C against the **production plan**; reviewed the data for the other groups.

• Compared actual vs standard consumption and actual vs planned production in Excel, and tracked downtime on six machines; prepared **supplier comparisons**, **reorder proposals** and weekly inventory, production and purchasing reports (**approximately 50%** less report-preparation time, operational estimate).

EDUCATION

MSc in Stochastics and Data Science · University of Turin | in progress, expected Apr 2027

Graduate studies in Engineering Management · Azad University, Iran | 2019–2021, program not completed (no degree awarded)

BSc in Industrial Engineering · Azad University, Iran | foreign degree, Oct 2014 – Jun 2019

SELECTED PROJECTS

Industrial Sensor Analytics and LSTM Anomaly Detection

MSc industry thesis, confidential real data; public synthetic demonstration

Problem: Analyze HVAC operating patterns and flag potential anomalies without fault labels.

Method: Python pipeline for 30-minute BMS/weather series, feature engineering, two seasonal LSTM autoencoders and Optuna tuning.

Output: Test-set reconstruction R^2 0.972 (summer) and 0.950 (winter); produced technical indicators. R^2 does not measure fault-detection accuracy.

Telecom Customer Churn and Segmentation

University project, real public dataset

Problem: Explore customer behavior, unusual usage, and churn likelihood.

Method: Analyzed 3,150 records with 13 variables using K-Means, PCA, Local Outlier Factor, and a Decision Tree classifier.

Output: Evaluated the classifier using accuracy, precision, and recall, accounting for the imbalanced churn class; no unverified accuracy percentage.

Steel Manufacturing Operations Analytics

Personal project, synthetic data (not real company data)

Problem: Identify potential capacity constraints, downtime concentration, and quality losses.

Method: Analyzed 3,780 simulated machine-shift records across seven machines; calculated OEE, plan vs. actual, scrap, downtime Pareto, and cycle times.

Output: Produced seven notebooks, KPI exports, charts and a summary of potential capacity constraints.

Demand forecasting · Personal portfolio (synthetic data)

Compared Holt-Winters with a seasonal naive baseline over 12 test months: MAPE of 2.79% versus 8.50%, a 67.2% relative error reduction.

LANGUAGES AND RECOGNITION

Languages: Italian B2 · English C1 · Armenian native · Persian professional

Awards (2025): 1st place (team) in the Michelin Students Green Challenge and the UniTo–SKF Students Indigo Challenge.

PRIVACY

I authorize the processing of my personal data for recruitment purposes under Regulation (EU) 2016/679 (GDPR) and applicable Italian legislation.